



EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Mid-Assessment, Summer 2025 Semester

Course: Electronic Circuits (CSE251/ICE213), Section:4
Instructor: Dr. Sarwar Jahan, Associate Professor, CSE Department
Full Marks: 30
Time: 1 Hour 20 Minutes

Note: There are 5 (Five) questions. Answer ALL of them. The Course outcome (CO), Cognitive level, and Mark of each question are mentioned in the right margin.

1. Find the currents and the voltages of each branch for the circuit of **Figure 1**. Obtain the voltage V_a . Considering the resistors R_2 and R_3 as a load, draw the load line and determine I_{DQ} and V_{DQ} . Sketch the waveform for the voltage across the load if the source voltage V_1 is a 60Hz sinusoid of 15V peak amplitude. Assuming all the diodes as Ge diodes. [CO1, C2, C3 Mark: 7]

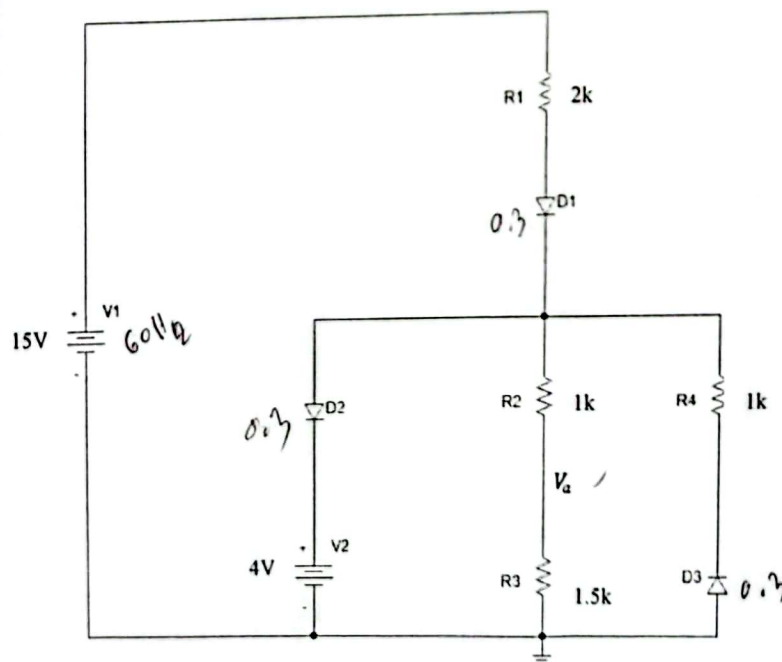


Figure 1

2. For the circuit of **Figure 2**, Determine the output voltage across the R_1 resistor, considering it as a load, and draw the output waveform. Find the Peak Inverse Voltage across the D_3 diode. Considering all the diodes as Si diodes. Identify the type of circuit represented in the given diagram. [CO2, C4, Mark: 5]

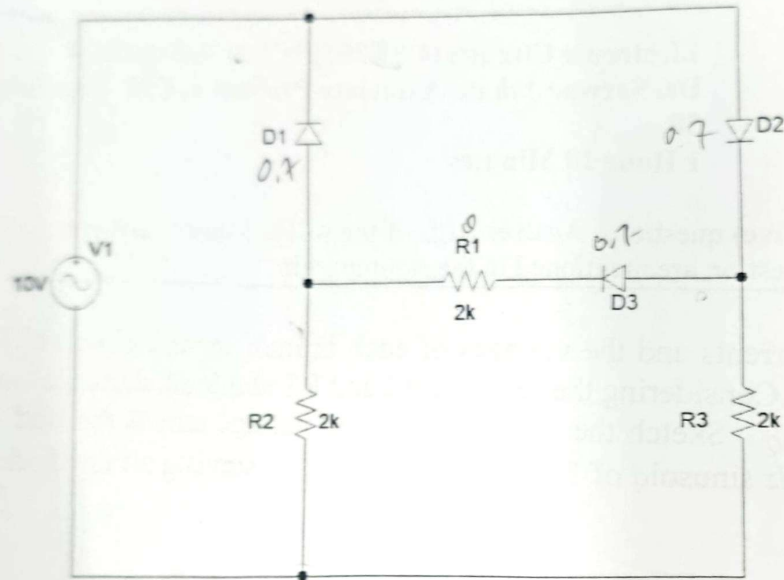


Figure 2

3. Design the network of the following Figure 3 to maintain V_L at 12 V for a load variation (I_L) from 0 mA to 200 mA. That is, determine R_S and V_Z and determine P_{Zmax} for the Zener diode. [CO1, C2, C3, Mark: 6]

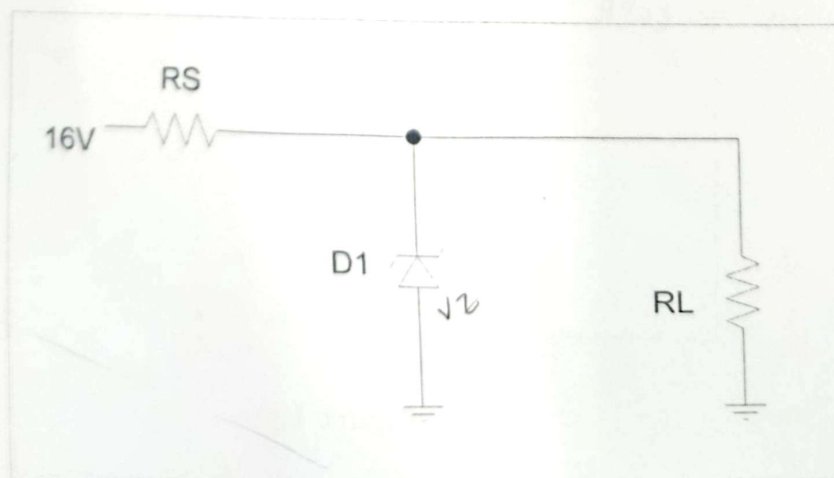


Figure 3

4. Given that $I_C = 2 \text{ mA}$ and $V_{CE} = 10 \text{ V}$, determine R_1 and R_C for the network of **Figure 4**. Show the dc load line analysis also. [CO1, CO2, C2 C3, Mark: 8]

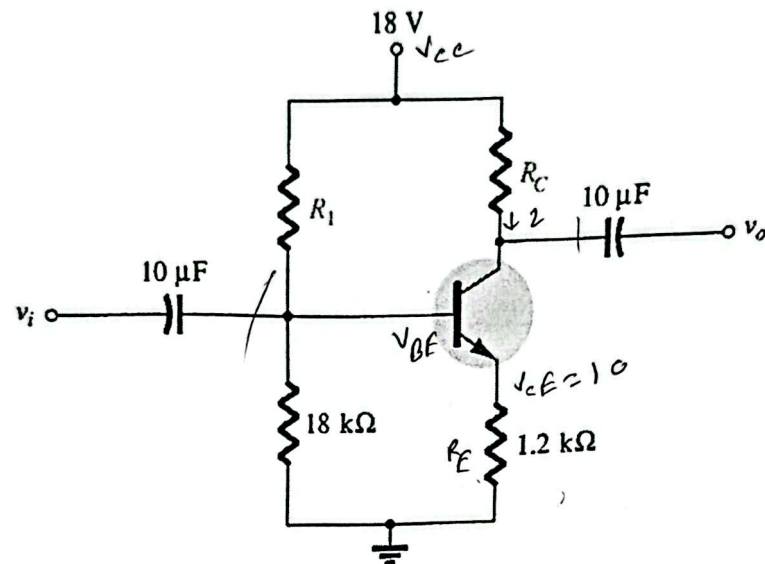


Figure 4

5. From the following circuit of **Figure 5**, determine the base current, the emitter current, collector current I_o , and the voltage V_{CE} . [CO1, C2 C3, Mark: 6]

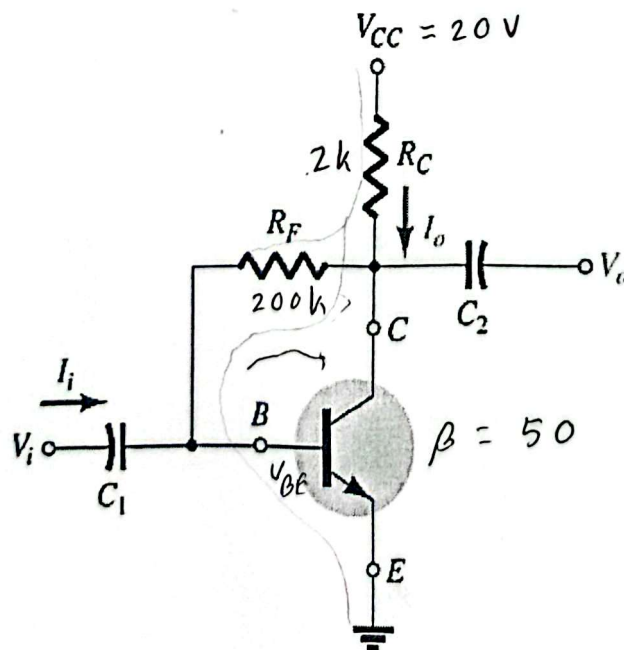


Figure. 5

Name: [Redacted]

ID: [Redacted]



Department of Computer Science and Engineering
Electronic Circuits (CSE251/ICE213); Section: 4
Course instructor: SJ
Quiz 1 (Summer 2025)

total: 15 Time: 25 mins

1. From the circuit of **Figure 1**, determine the currents and voltages in each branch. If $10\sin\omega t$ is applied instead of 10V dc show the output voltage wave form across the R3 resistor. Consider all as Si diode. [CO1, Mark: 15]

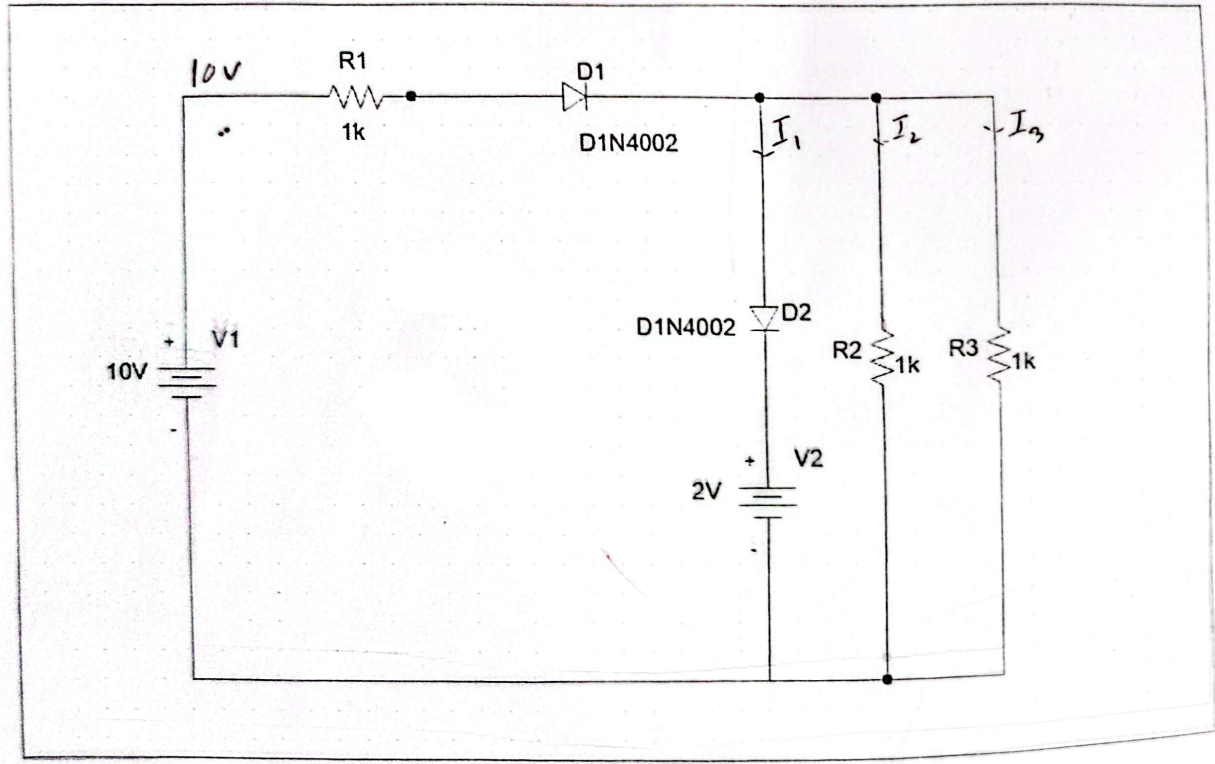


Figure 1

Time: 25 mins

1. Find V_o and i_o in the circuit of Figure 1.

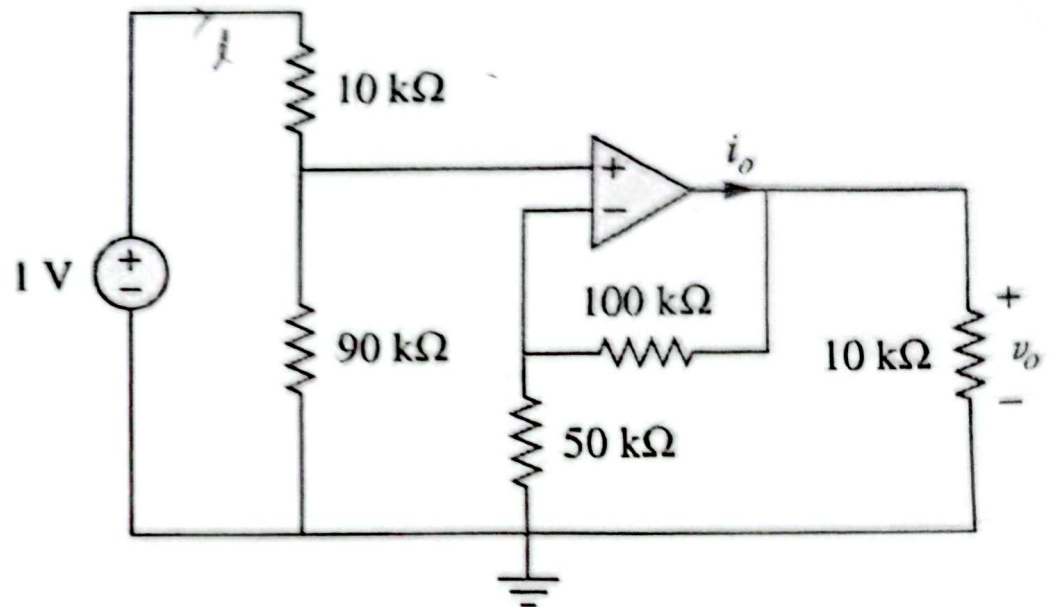


Figure 1

2. If $V_1 = 7V$ and $V_2 = 3.1V$ find V_o in the op amp circuit of Figure 2.

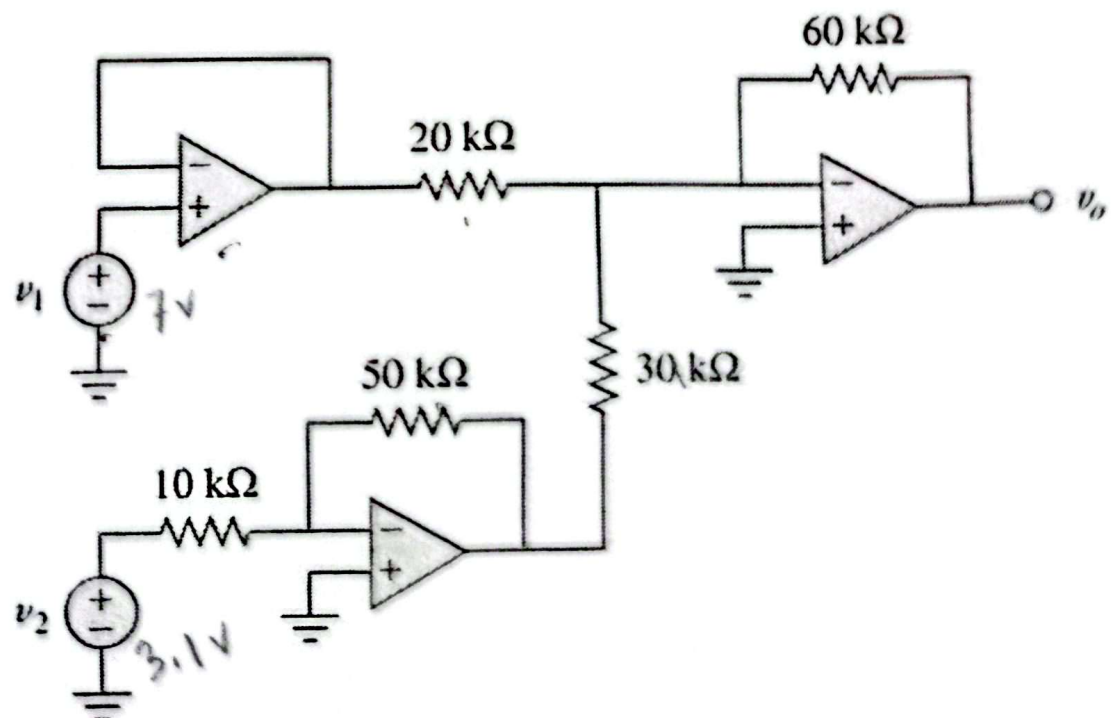


Figure 2